Starjot

Personal Project 04

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2 November 2026

Table of Contents

1. Overview 3
1.1 Problem 3
1.2 Solution 3
1.3 Components 3
1.4 Features ..... **Error! Bookmark not defined.**
2. Scope 4
2.1 In Scope 4
2.2 Out of Scope 4
3. Setup & Design 4
4. Methodology 5
5. Limitations & Risks 6
6. Testings and Findings 6
7. Challenges 6
8. Improvements 7

1. Overview

Starjot is a browser-based workspace for creating notes and organising project work in one place.

It combines a note editor and canvas with project views, helping users capture information, organise ideas, and track work without switching between separate tools.

1.1 Problem

Notes, ideas, and project tasks can become scattered across different tools. This can make it harder to find information, connect related notes, and keep track of ongoing work.

1.2 Solution

Starjot brings notes and project organisation together in a single workspace. Users can create and organise notes, capture ideas on a canvas, and manage project work through different views.

1.3 Goals

- Make it easy to capture and organise information
- Keep notes and project work accessible in one workspace
- Support different ways of viewing and managing work
- Allow users to find and connect related information
- Save workspace data locally in the browser

1.4 Features

Feature	Description
Notespace	Create, edit, organise, and search notes.
Rich Text Editing	Format note content and work with blocks, links, images, and tables.
Canvas	Arrange ideas visually using canvas boxes and drawings.
Categories & Organisation	Group notes into categories, star notes, and use different note views.
Links & Backlinks	Connect related content and find references between notes.
Import & Export	Import supported files and export notes or canvas content.
Boardspace	Organise project work using summary, list, board, and timeline views.

Reminders & Inbox	Set note reminders and review notifications in the inbox.
Search	Find information across the workspace.
Local Storage	Save workspace data and preferences in the browser.

2. Scope

2.1 In Scope

- Creating, editing, and organising notes
- Rich text, images, tables, links, and drawings
- Canvas-based idea organisation
- Categories, starred notes, and search
- Importing and exporting supported content
- Project organisation through summary, list, board, and timeline views
- Reminders and an inbox
- Locally saved workspace data and preferences

2.2 Out of Scope

The current implementation does not provide:

- Cloud accounts or cross-device synchronisation
- Real-time collaboration between users
- A server-hosted workspace

3. Setup & Design

3.1 Technology Stack

Starjot is a client-side web application built with:

- React and TypeScript for the application interface and behaviour
- CSS for styling
- Vite for development and building
- Browser storage for locally saved data
- Vitest for automated testing

3.2 System Architecture

Starjot separates the user interface, shared data types, and browser storage into different parts of the project.

- Notespace handles notes, editing, canvas content, and note organisation.
- Boardspace handles project work and its different views.
- Storage modules read and save workspace data and preferences in the browser.
- Search and shared components support navigation and interaction across the application.

3.3 Design Decisions

- Notes and project work are placed in the same application so related information is easier to access.
- Project work can be viewed as a list, board, or timeline depending on the user's needs.
- Notes can contain structured content as well as visual canvas elements.
- Workspace data is stored locally, allowing the application to work without a server-hosted account.
- Storage logic is separated from the interface to make it easier to maintain or replace later.

3.4 Running the Project

Install the project dependencies with npm install, then run npm run dev to start the development server.

4. Methodology

Starjot was developed as a modular application, with separate areas for notes, project work, storage, and shared interface components.

Features are supported by automated tests covering application behaviour, storage, editing, import and export, and other workspace functions. The project also includes type checking and a production build as part of its checks.

5. Limitations & Risks

- Workspace data is stored in the current browser and may be lost if browser storage is cleared or becomes unavailable.
- Local data is not automatically synchronised between devices.
- Browser storage capacity can limit the amount of data that can be saved.
- Imported content may not retain every detail of its original formatting.
- Reminders depend on the application and browser environment; they should not be treated as a guaranteed notification service.
- Complex editing and canvas content may behave differently across browsers or screen sizes.

6. Testings and Findings

Starjot includes automated tests for areas such as note editing, storage, search, reminders, canvas, import and export, and project views.

- Run `npm test` to run the automated test suite.
- Run `npm run` check to run type checking, tests, and the production build.

The automated tests provide repeatable checks, but they do not replace manual review of the interface and browser-specific behaviour.

7. Challenges

Challenge	How it was handled
Supporting several kinds of note content	The editor separates functionality such as text, images, tables, links, and drawings into modules.
Presenting project work in different ways	Boardspace provides summary, list, board, and timeline views of project work.
Keeping locally saved data usable	Storage modules handle reading and writing workspace data and retain copies of damaged data where possible.

8. Improvements

- **Desktop Application**
  - Develop an installable Windows application, with macOS support planned afterwards.
  - Adapt and test the editor, window controls, shortcuts, and printing for desktop use.
- **Workspace Storage**
  - Store workspace data in a user-selected local folder instead of relying on browser storage.
  - Support moving an existing browser workspace into the desktop application.
- **Backup & Recovery**
  - Add a full workspace backup and restore option, including notes, images, and project data.
  - Improve recovery from damaged files, failed saves, and conflicting changes.
- **Cross-Device Access**
  - Design workspace folders to work with third-party services such as OneDrive or Google Drive.
  - Detect conflicts and protect user data when changes occur on different devices.
- **Canvas Improvements**
  - Add more visual elements, such as images, shapes, and sticky notes.
- **Attachments**
  - Support additional files, such as PDFs, Word documents, spreadsheets, and ZIP files.
  - Explore in-app previews and opening files in external applications.
- **Notifications**
  - Explore desktop notifications and reminders when Starjot is not open.
- **Accessibility & Compatibility**
  - Continue improving keyboard navigation and accessibility.
  - Expand testing across browsers, screen sizes, and future desktop platforms.
- **Mobile Application**
  - Explore a mobile version after the desktop application is stable.